Algorithms & Data Structure In Python

Python Data Types

Lab_No-1

Basics of Python Programming

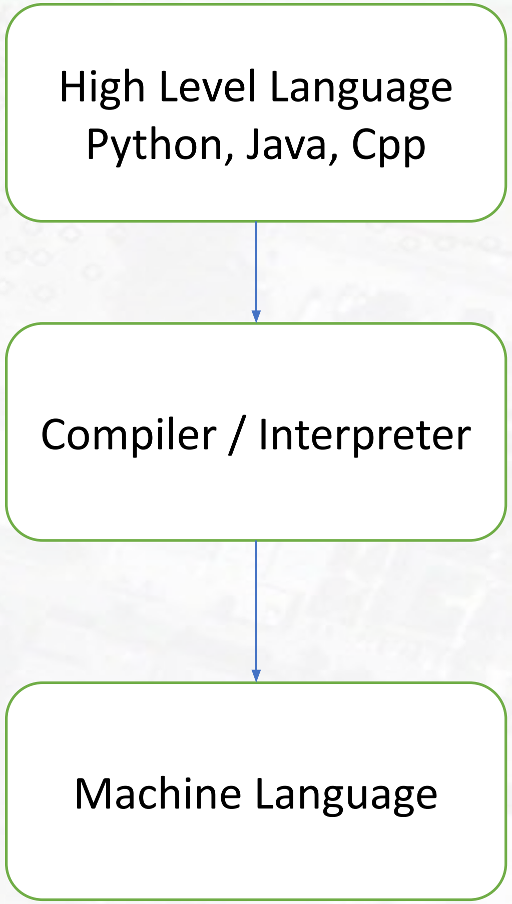
2020

Programming Languages

- **High Level Programming Language**: : Python, Java, C++, JavaScript
- **Assembly language**: A low-level symbolic representation of machine instructions.
- **Machine language**: Binary instructions executed directly by the processor: 0s and 1s.

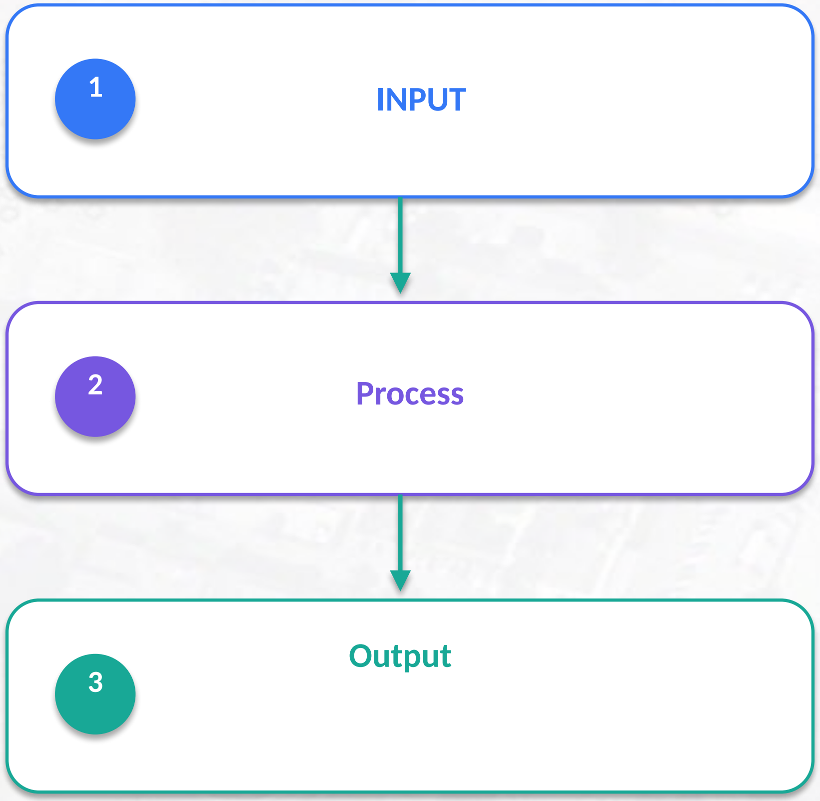
Modern **AI** uses **Natural Language Processing** (NLP) to enable computers to analyze and generate human language.

Programs Execution



Algorithm Structures

- Sequence
- Decision
- Repetition



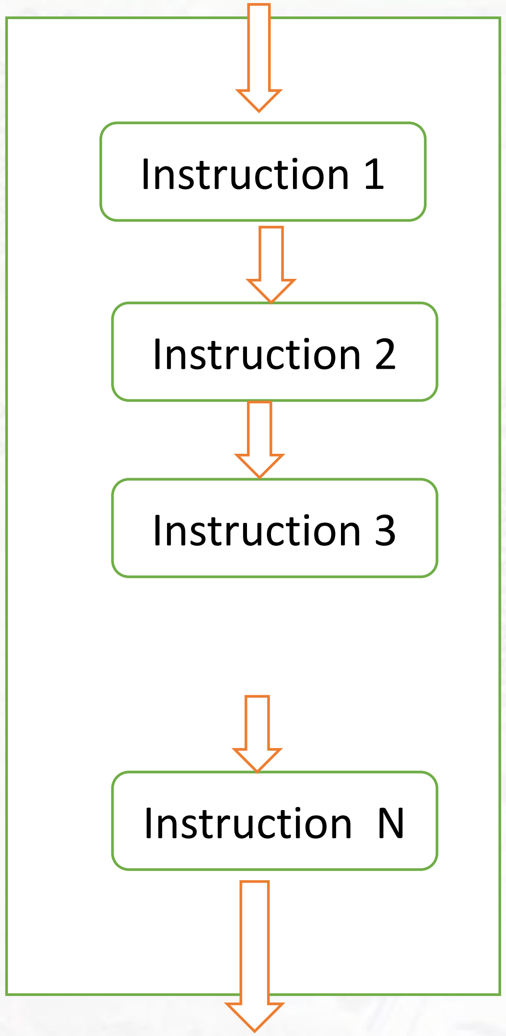
Representation of Algorithms

- **UML (Unified Modeling Language)**
  - A **visual way to represent** the structure and flow of an algorithm or system.
  - Uses diagrams to show the **main steps, decisions, actions, and relationships**.
  - Helps learners understand the **big picture** without focusing on programming-language syntax.

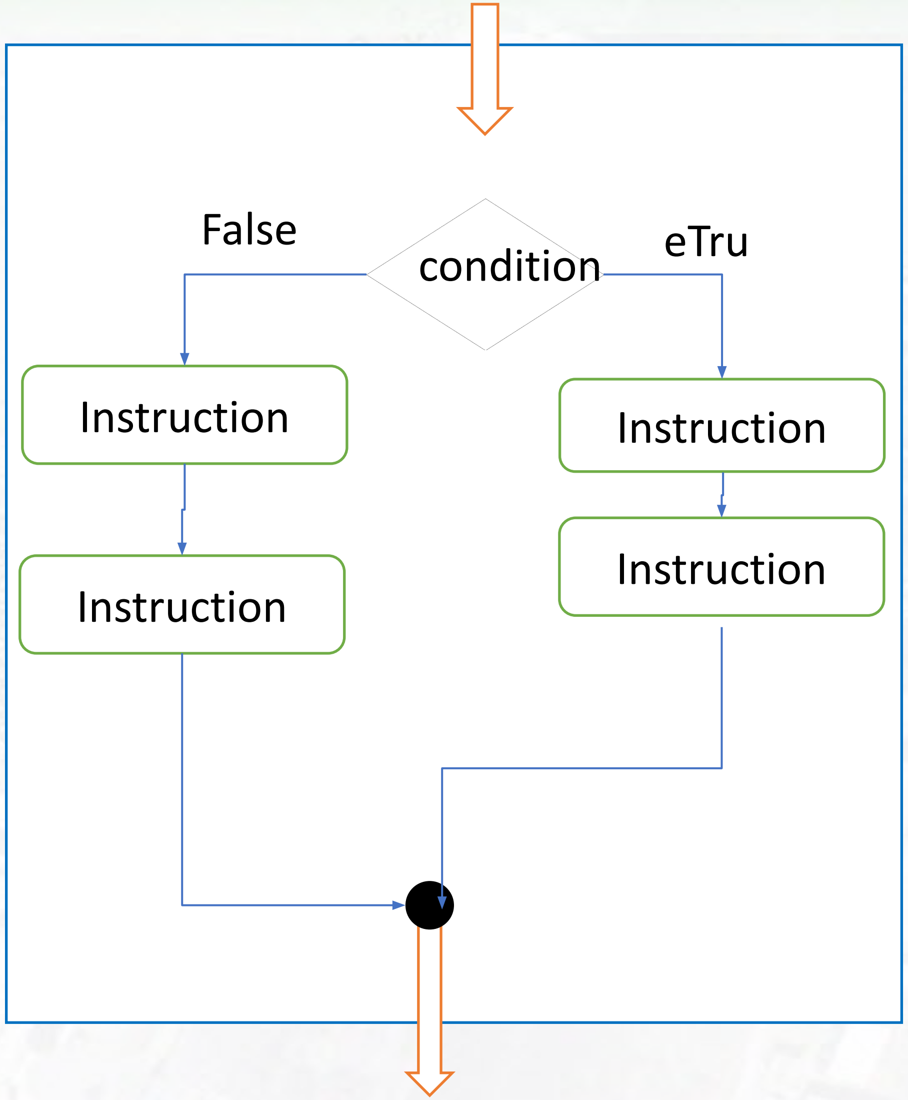
- **Pseudocode**
  - A **text-based description** of an algorithm written in simple, structured language.
  - Expresses the algorithm as **steps, decisions, loops, inputs, and outputs**.
  - Is **independent of any programming language**.

UML Algorithm Representation

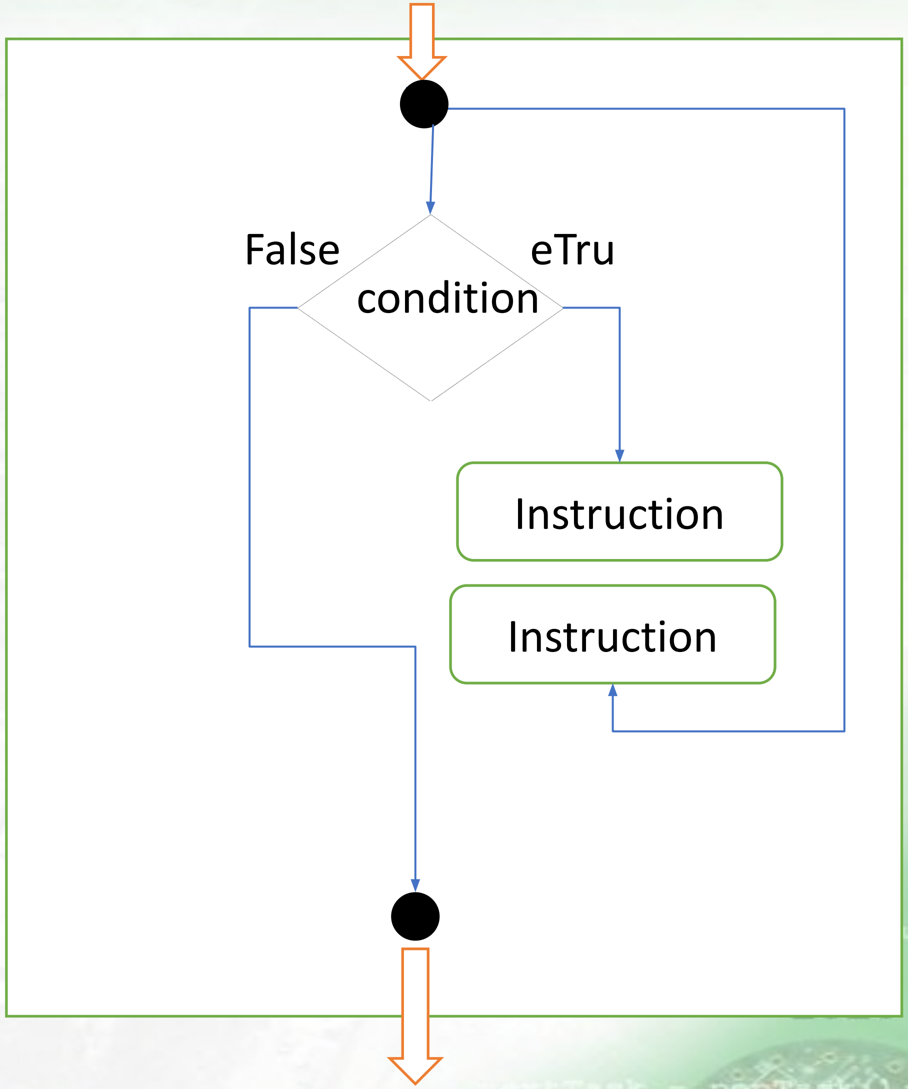
Sequential Structures



Decision Structure



Repetition Structures of



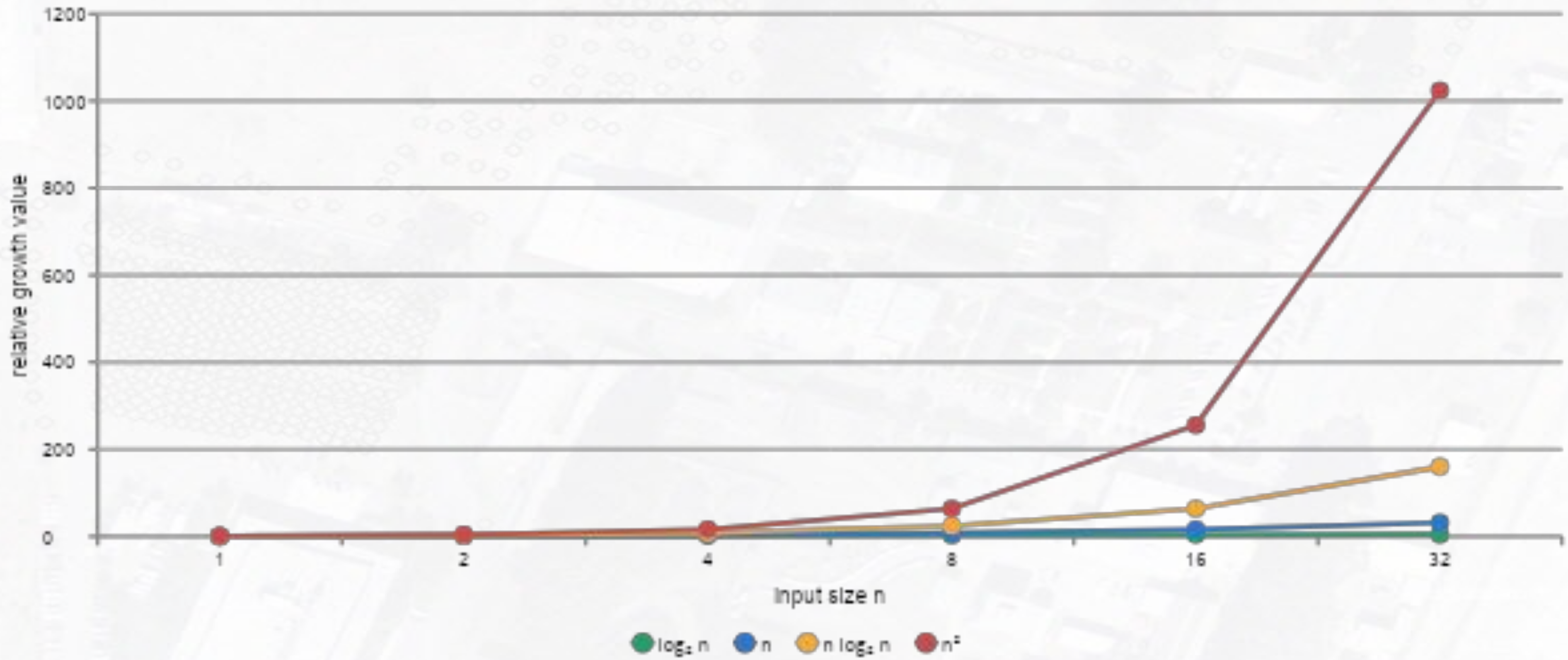
Pseudo Code Algorithm Representation

```
INPUT unit_price, quantity

REQUIRE unit_price ≥ 0
REQUIRE quantity ≥ 0

total ← unit_price × quantity

OUTPUT total
```

Predict first: unit_price = 12.5 and quantity = 3 → total = ?

Big-O Notation

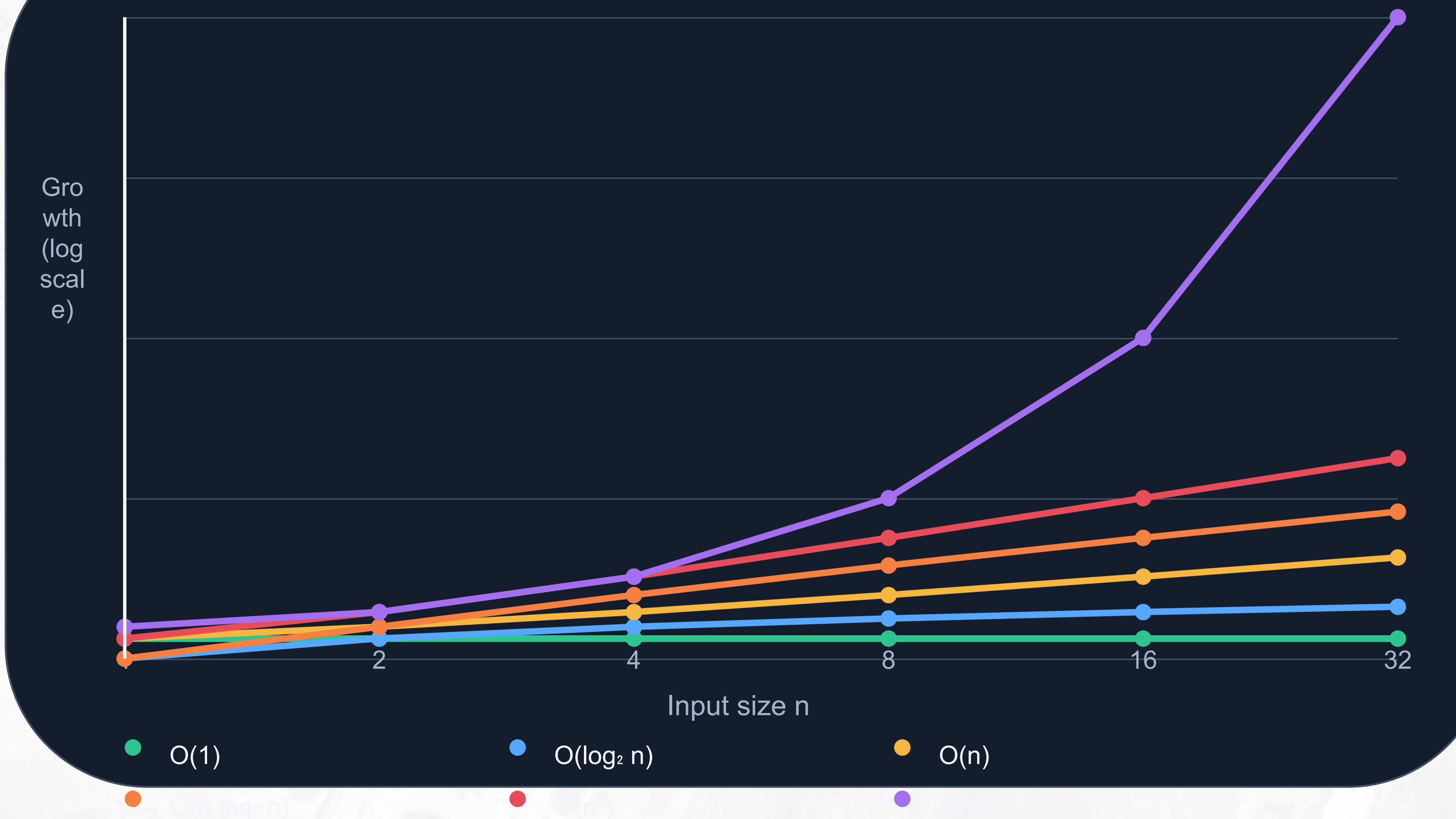
- **Big-O notation** describes how an algorithm's **time or memory requirements grow as the input size** n **increases**.



At n = 32, $n^2 = 1024$ while $n \log_2 n = 160$.

Big-O Growth: How Algorithms Scale

1 | 2 | 4 | 8 | 16 | 32



At n = 1

Representative growth values

O(1)	1
$O(\log_2 n)$	0
O(n)	1
$O(n \log_2 n)$	0
$O(n^2)$	1
$O(2^n)$	2

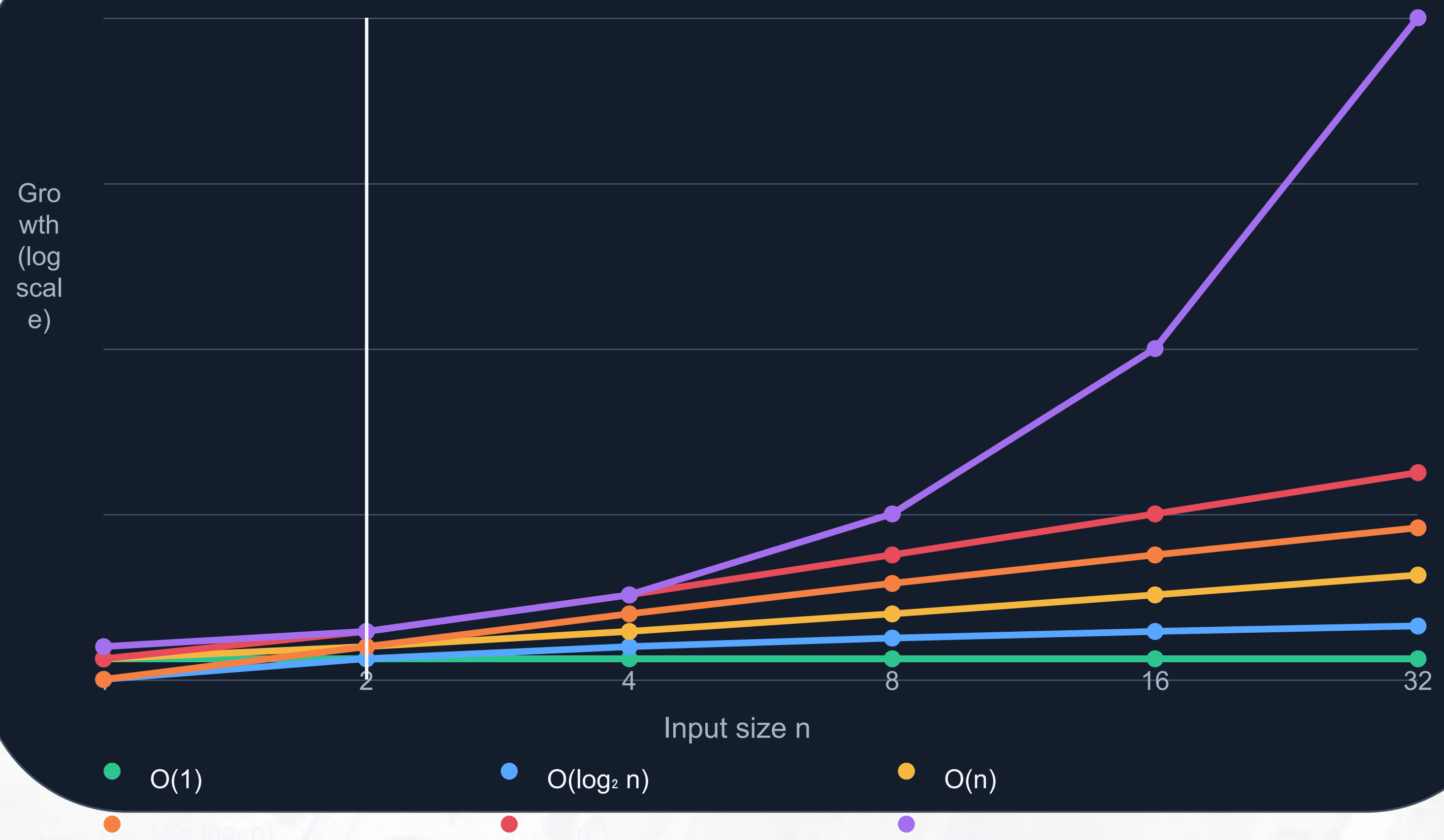
Key idea Different growth rates separate dramatically as n increases.

Values illustrate growth functions, not measured execution time.

Big-O Growth: How Algorithms Scale

Select an input size n to compare representative growth values.

1 | 2 | 4 | 8 | 16 | 32



At n = 2

Representative growth values

O(1)	1
O($\log_2$ n)	1
O(n)	2
O(n $\log_2$ n)	2
O(n^2)	4
O(2^n)	4

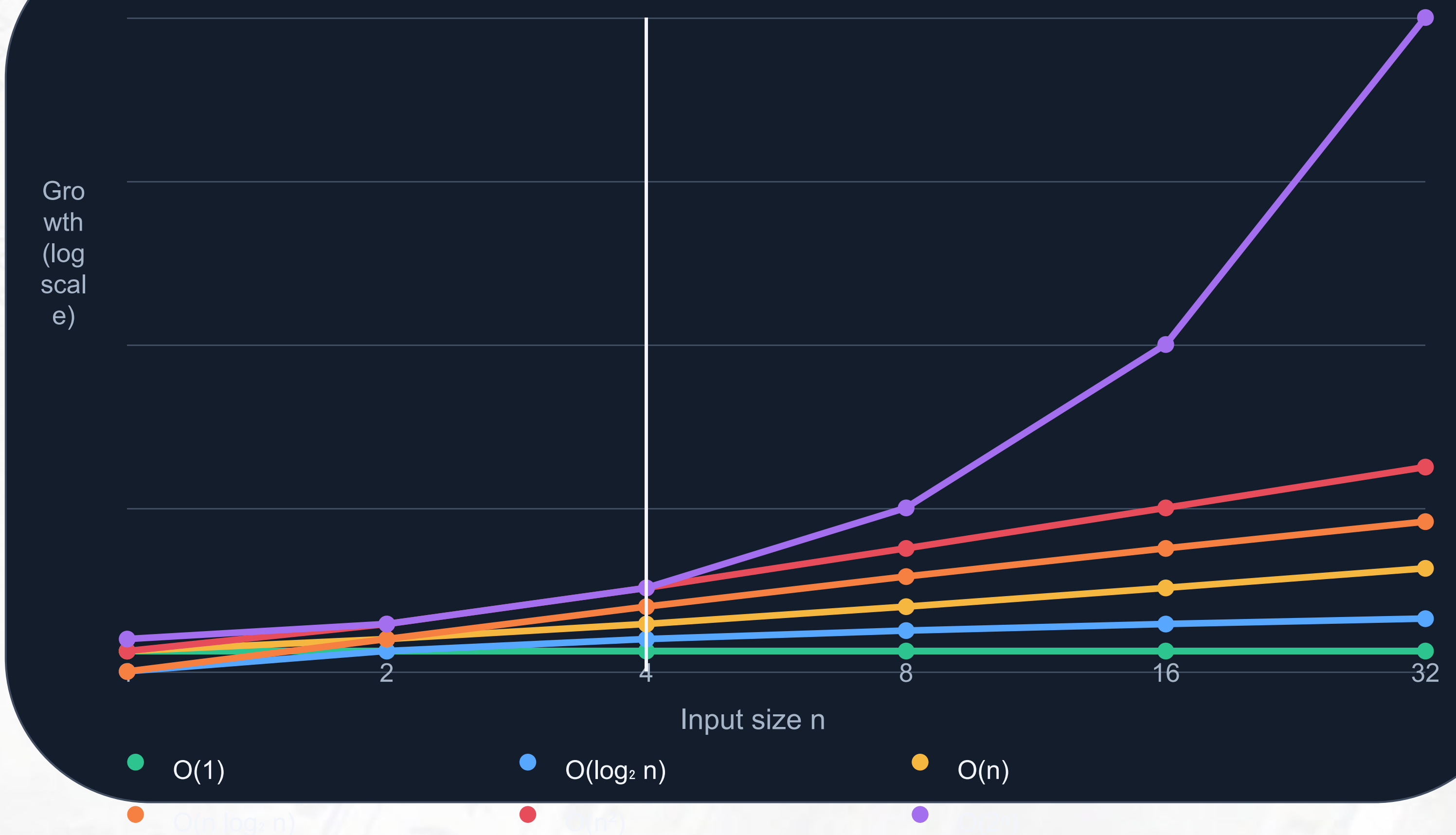
Key idea Different growth rates separate dramatically as n increases.

Values illustrate growth functions, not measured execution time.

Big-O Growth: How Algorithms Scale

Select an input size n to compare representative growth values.

1 | 2 | 4 | 8 | 16 | 32



At n = 4

Representative growth values

O(1)	1
O($\log_2$ n)	2
O(n)	4
O(n $\log_2$ n)	8
O(n^2)	16
O(2^n)	16

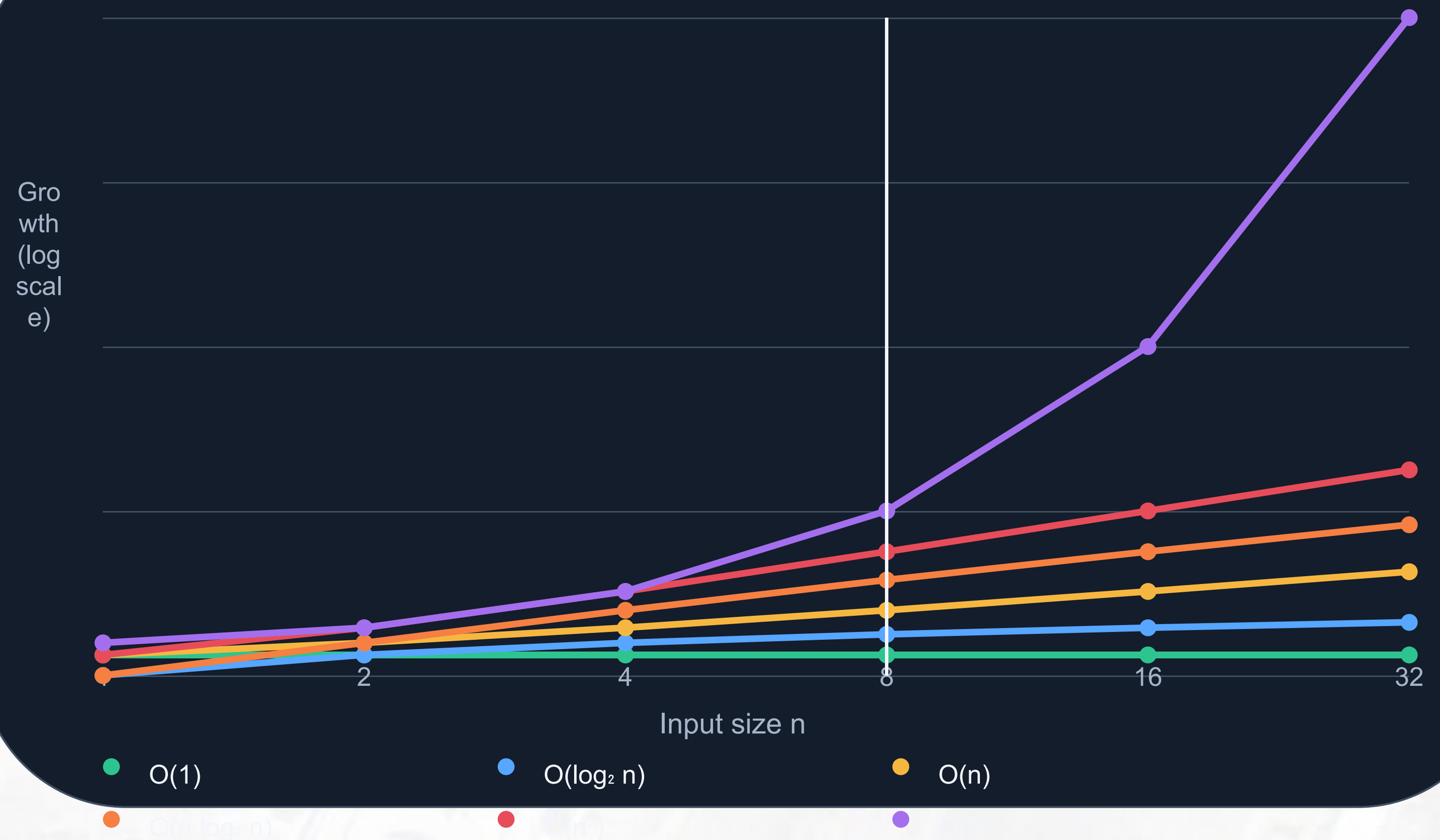
Key idea Different growth rates separate dramatically as n increases.

Values illustrate growth functions, not measured execution time.

Big-O Growth: How Algorithms Scale

Select an input size n to compare representative growth values.

1 | 2 | 4 | **8** | 16 | 32



At n = 8

Representative growth values

O(1)	1
$O(\log_2 n)$	3
O(n)	8
$O(n \log_2 n)$	24
$O(n^2)$	64
$O(2^n)$	256

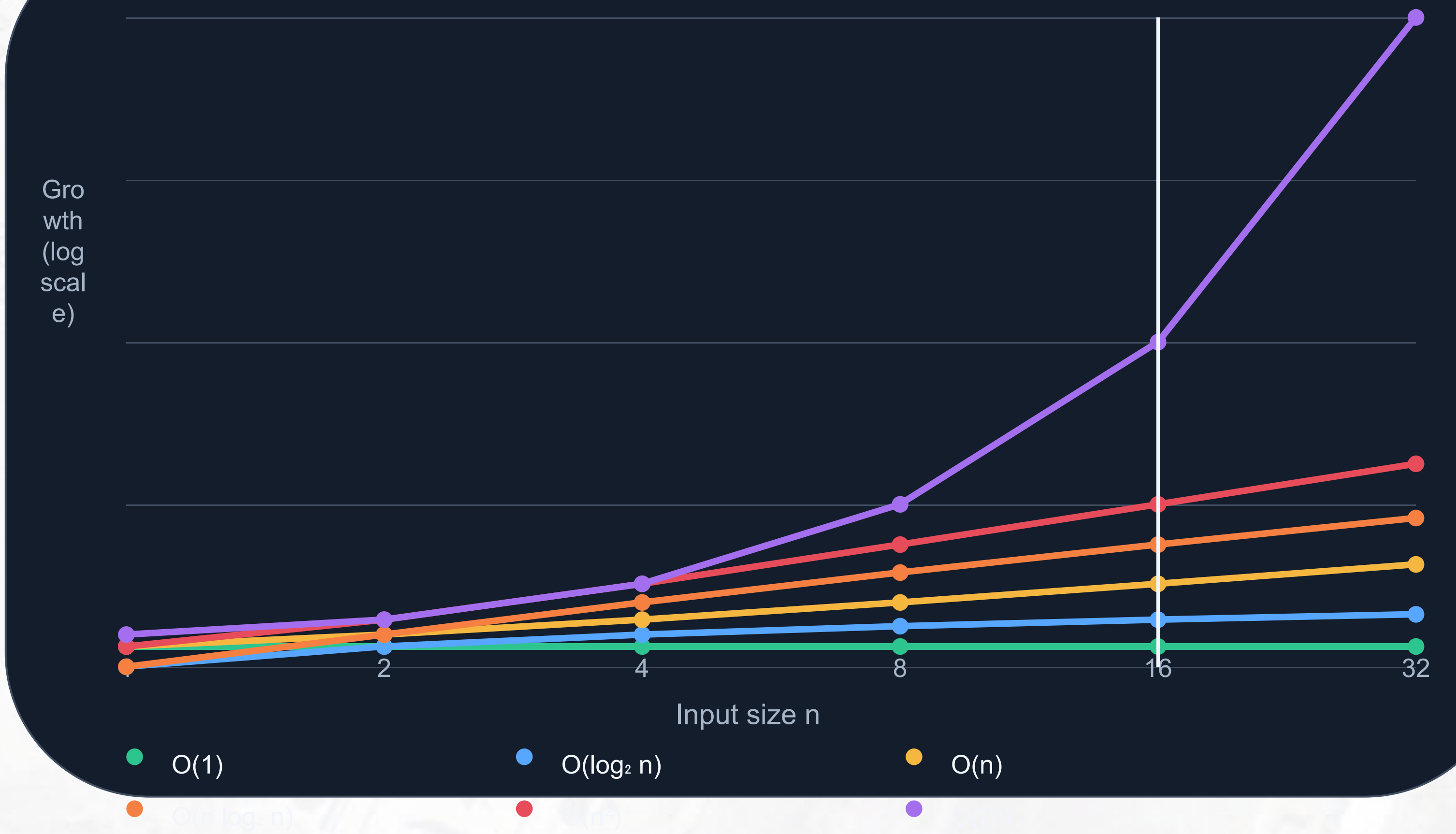
Key idea Different growth rates separate dramatically as n increases.

Values illustrate growth functions, not measured execution time.

Big-O Growth: How Algorithms Scale

Select an input size n to compare representative growth values.

1 | 2 | 4 | 8 | **16** | 32



At n = 16

Representative growth values

O(1)	1
O($\log_2 n$)	4
O(n)	16
O($n \log_2 n$)	64
O(n^2)	256
O(2^n)	65,536

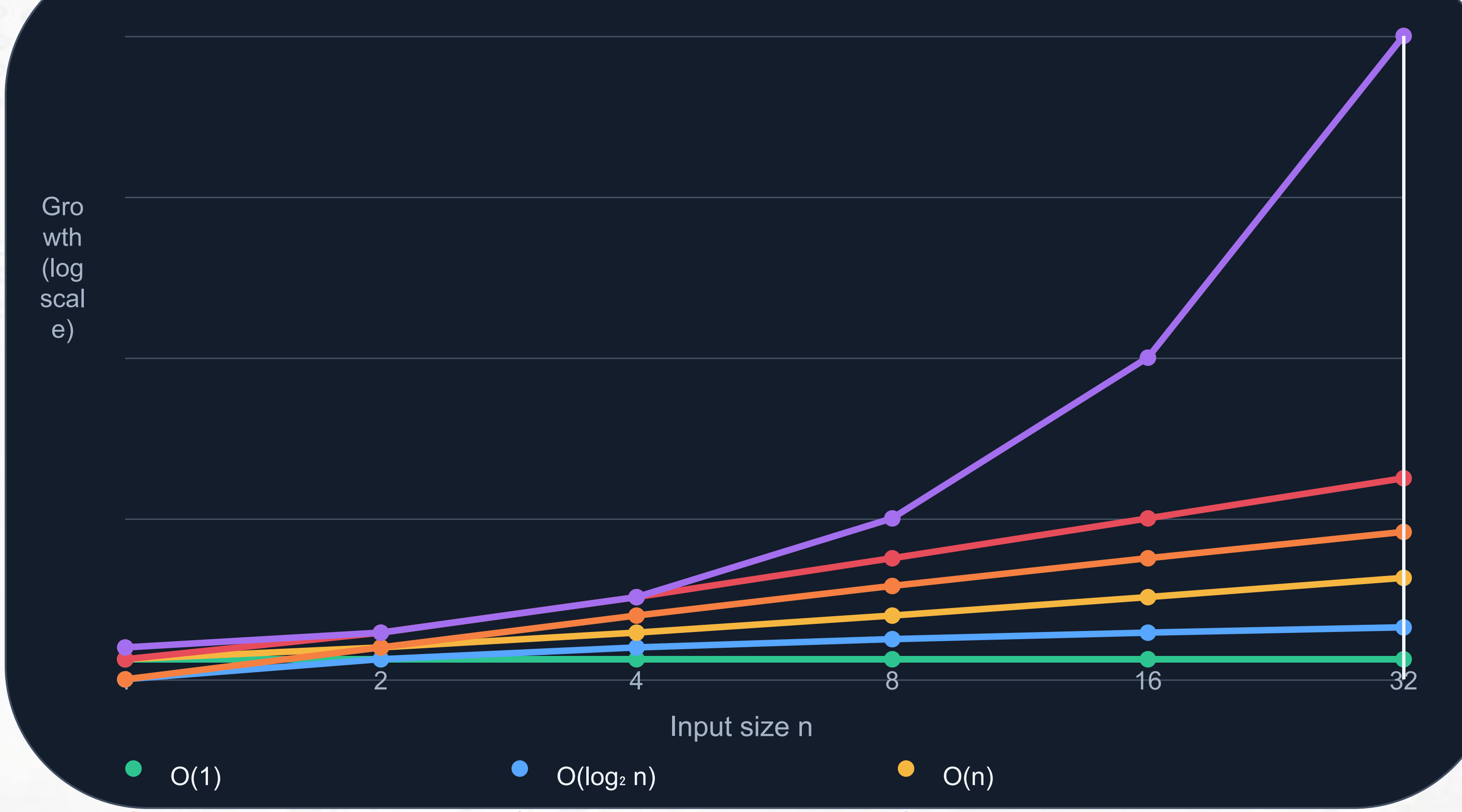
Key idea Different growth rates separate dramatically as n increases.

Values illustrate growth functions, not measured execution time.

Big-O Growth: How Algorithms Scale

Select an input size n to compare representative growth values.

1 | 2 | 4 | 8 | 16 | **32**



At n = 32

Representative growth values

O(1)	1
$O(\log_2 n)$	5
O(n)	32
$O(n \log_2 n)$	160
$O(n^2)$	1,024
$O(2^n)$	4,294,967,296

Key idea Different growth rates separate dramatically as n increases.

Values illustrate growth functions, not measured execution time.

Search illustrates how data organization changes the algorithm

Linear search works on arbitrary sequences; binary search requires sorted data and repeatedly halves the remaining search interval.

Target = 19

Linear search

2	5	8	12	19	27	31

checks from left to right → 5 comparisons here • worst case O(n)

Binary search

2	5	8	12	19	27	31

1st middle: 12 → go right

2nd middle: 27 → go left

3rd: 19 → found

halves the remaining interval • worst case O(log n)

PYTHON

```
def linear_search(a, target):
    for i, value in enumerate(a):
        if value == target:
            return i
    return -1
```

Binary search is only correct when the data are sorted according to the same ordering used by the search.

From an Algorithm to Reliable Software

The same improvement cycle appears at every scale.

1. **Problem** — What must be solved?
2. **Algorithm** — Define the steps
3. **Python Code** — Implement the solution
4. **Test** — Check correctness
5. **Improve** — Learn and iterate

Agile repeat in small increments

CI automate build & tests

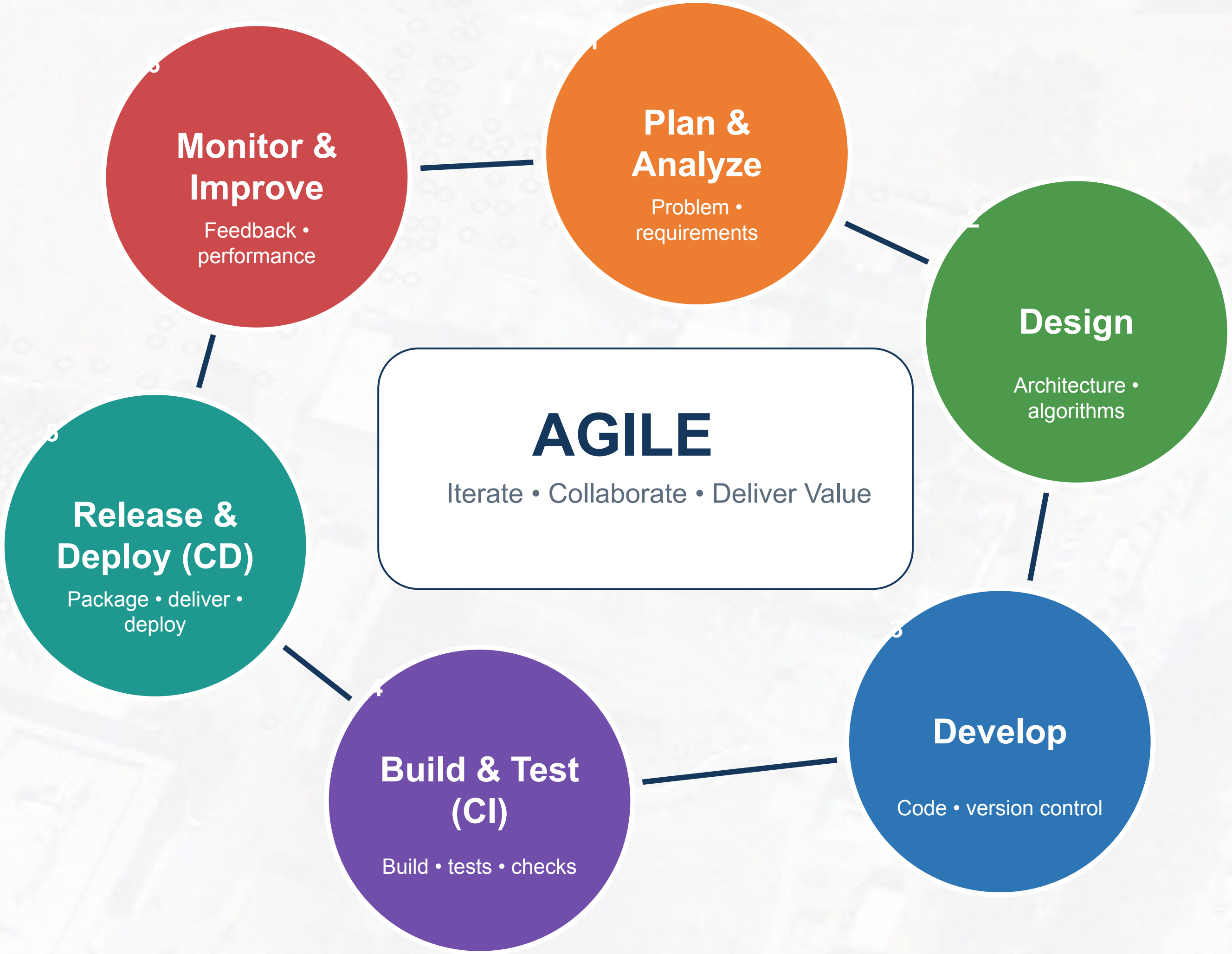
CD automate delivery/deployment

software development is a continuous learning cycle, not a one-time sequence.

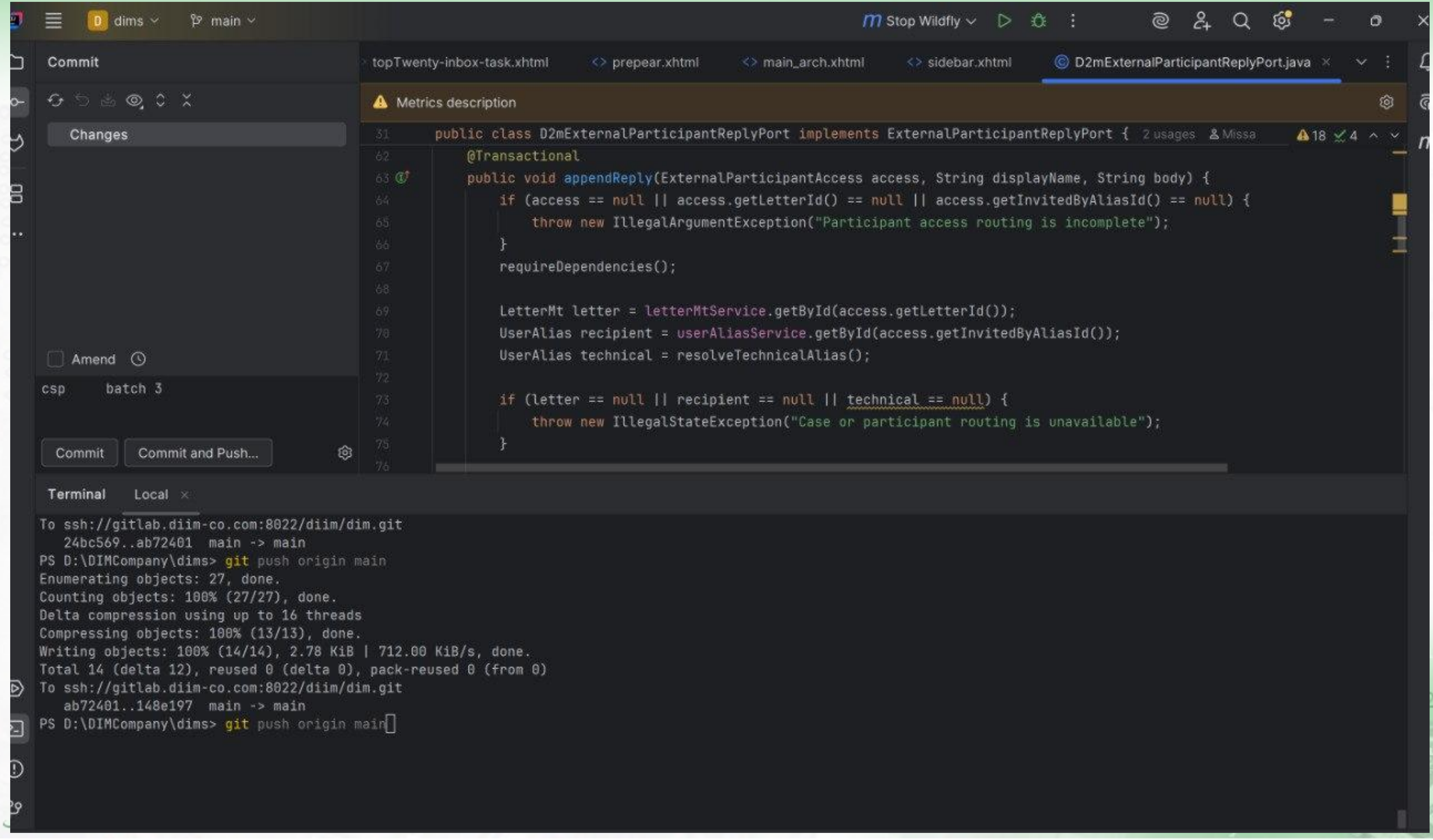
Application Development Cycle

Agile development with CI/CD

Build small → test early → deliver often → learn continuously



Problem → Algorithm → Code → Test → Improve


dims
main
Stop Wildfly
Commit
Changes
Amend
csp batch 3
Commit
Commit and Push...
topTwenty-inbox-task.xhtml
prepear.xhtml
main_arch.xhtml
sidebar.xhtml
D2mExternalParticipantReplyPort.java
Metrics description
public class D2mExternalParticipantReplyPort implements ExternalParticipantReplyPort {
2 usages
Missa
@Transactional
public void appendReply(ExternalParticipantAccess access, String displayName, String body) {
if (access == null || access.getLetterId() == null || access.getInvitedByAliasId() == null) {
throw new IllegalArgumentException("Participant access routing is incomplete");
}
requireDependencies();
LetterMt letter = letterMtService.getById(access.getLetterId());
UserAlias recipient = userAliasService.getById(access.getInvitedByAliasId());
UserAlias technical = resolveTechnicalAlias();
if (letter == null || recipient == null || technical == null) {
throw new IllegalStateException("Case or participant routing is unavailable");
}
Terminal
Local
To ssh://gitlab.diim-co.com:8022/diim/dim.git
24bc569..ab72401 main -> main
PS D:\DIMCompany\dims> git push origin main
Enumerating objects: 27, done.
Counting objects: 100% (27/27), done.
Delta compression using up to 16 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (14/14), 2.78 KiB | 712.00 KiB/s, done.
Total 14 (delta 12), reused 0 (delta 0), pack-reused 0 (from 0)
To ssh://gitlab.diim-co.com:8022/diim/dim.git
ab72401..148e197 main -> main
PS D:\DIMCompany\dims> git push origin main

Не защищено https://gitlab.diim-co.com/diim/dim

DIIM / dim

D dim

Star 0 Fork 0

Search or go to...

Project

D dim

Pinned

Issues 0

Merge requests 0

Manage

Plan

Code

Build

Secure

Deploy

Operate

Monitor

Analyze

Settings

Help

Admin Area

main dim / + History Find file Edit Code

csp batch 3
Missa authored 7 minutes ago

148e197d

Name	Last commit	Last update
src/main	csp batch 3	7 minutes ago
.gitattributes	Clean DIM Java source project	1 month ago
.gitignore	Clean DIM Java source project	1 month ago
Jenkinsfile	Add Jenkins pipeline for WAR de...	1 month ago
pom.xml	telegram wahts batch 5	2 weeks ago

Project information

582 Commits

1 Branch

0 Tags

14.9 MiB Project Storage

+ Add README
+ Add LICENSE
+ Add CHANGELOG
+ Add CONTRIBUTING
+ Enable Auto DevOps
+ Add Kubernetes cluster
+ Set up CI/CD
+ Add Wiki
+ Configure Integrations

Created on

July 04, 2026

gitlab-classroom.diim-co.com/company/apps/diim/dim/-/settings/ci_cd#autodevops-settings

Перезапустить и обновить

Search or go to...

0 3 1 Admin

Project

- Code
- Build
- Secure
- Deploy
- Operate
- Monitor
- Analyze
- Settings
  - General
  - Service accounts
  - Integrations
  - Webhooks
  - Access tokens
  - Repository
  - Merge requests
  - CI/CD
  - Packages and registries
  - Monitor
  - Usage quotas
- Help
- Collapse sidebar

Company Software / Applications / DIIM / dim / **CI/CD Settings**

Search page

General pipelines

Customize your pipeline configuration.

Auto DevOps

Automate building, testing, and deploying your applications based on your continuous integration and delivery configuration. How do I get started?

- [x] Default to Auto DevOps pipeline `instance enabled`

The Auto DevOps pipeline runs if no alternative CI configuration file is found. Learn more.

Deployment strategy

- (•) Continuous deployment to production
- ( ) Continuous deployment to production using timed incremental rollout
- ( ) Automatic deployment to staging, manual deployment to production

Save changes

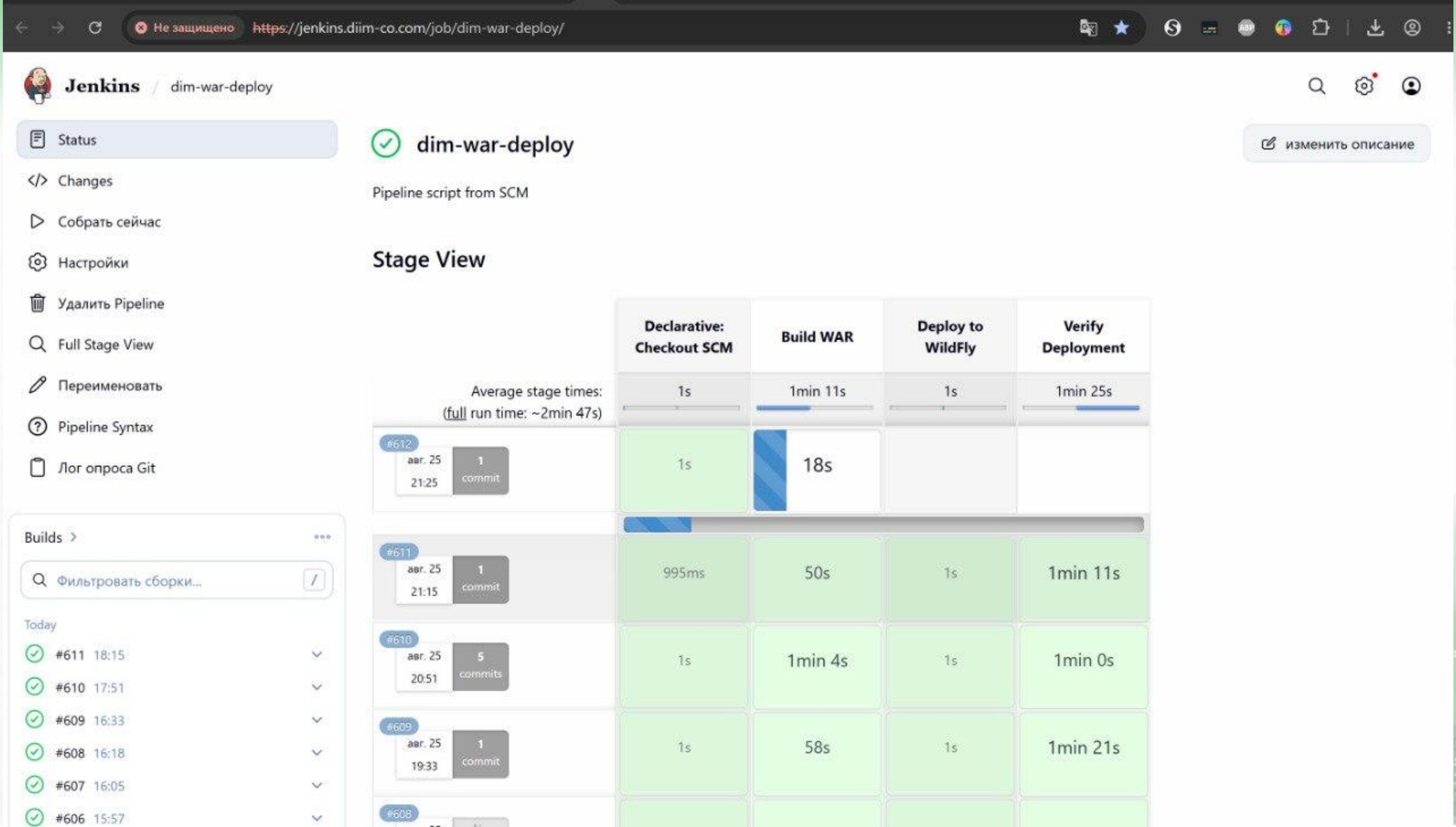
Runners

Runners are processes that pick up and execute CI/CD jobs for GitLab. What is GitLab Runner?

noun: бегун, бегунок, дорожка, ротор, полоз...

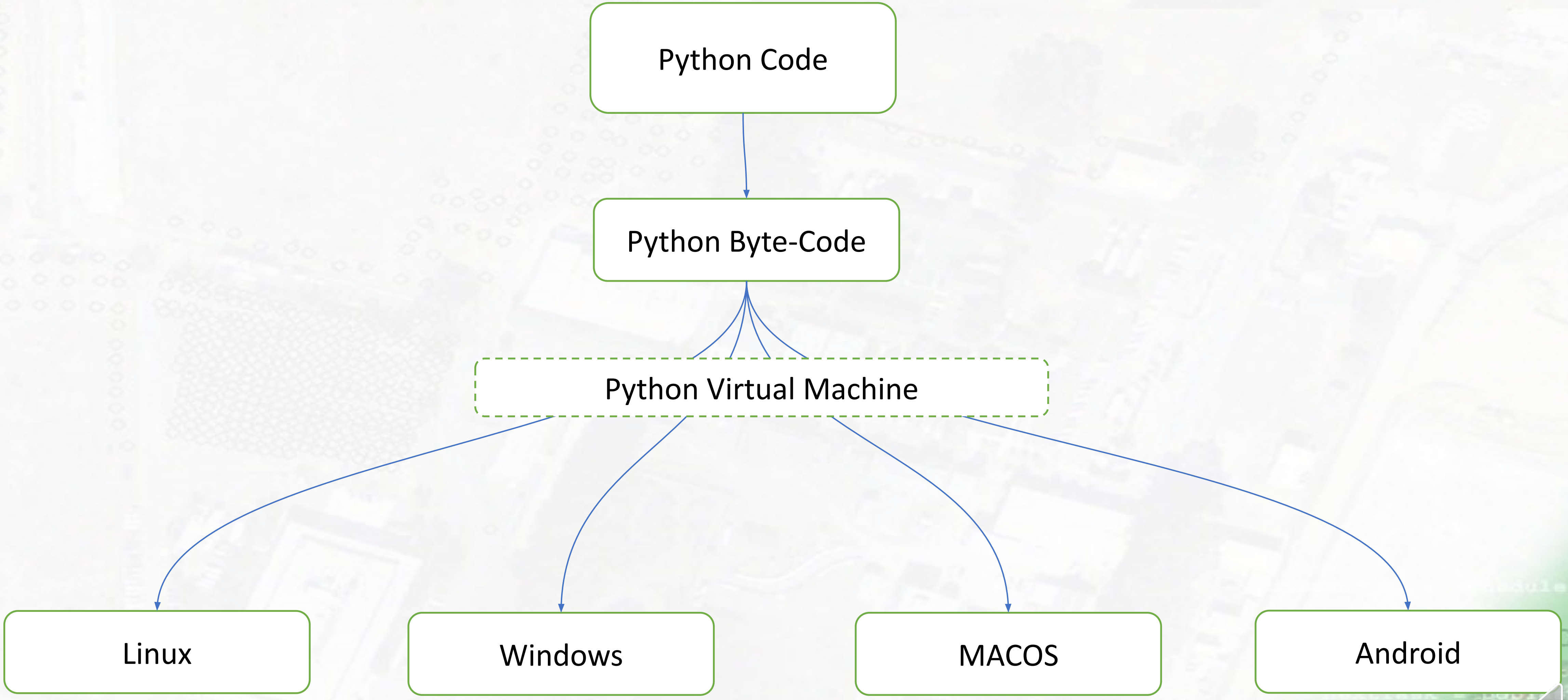
Artifacts

A job artifact is an archive of files and directories saved by a job when it finishes.

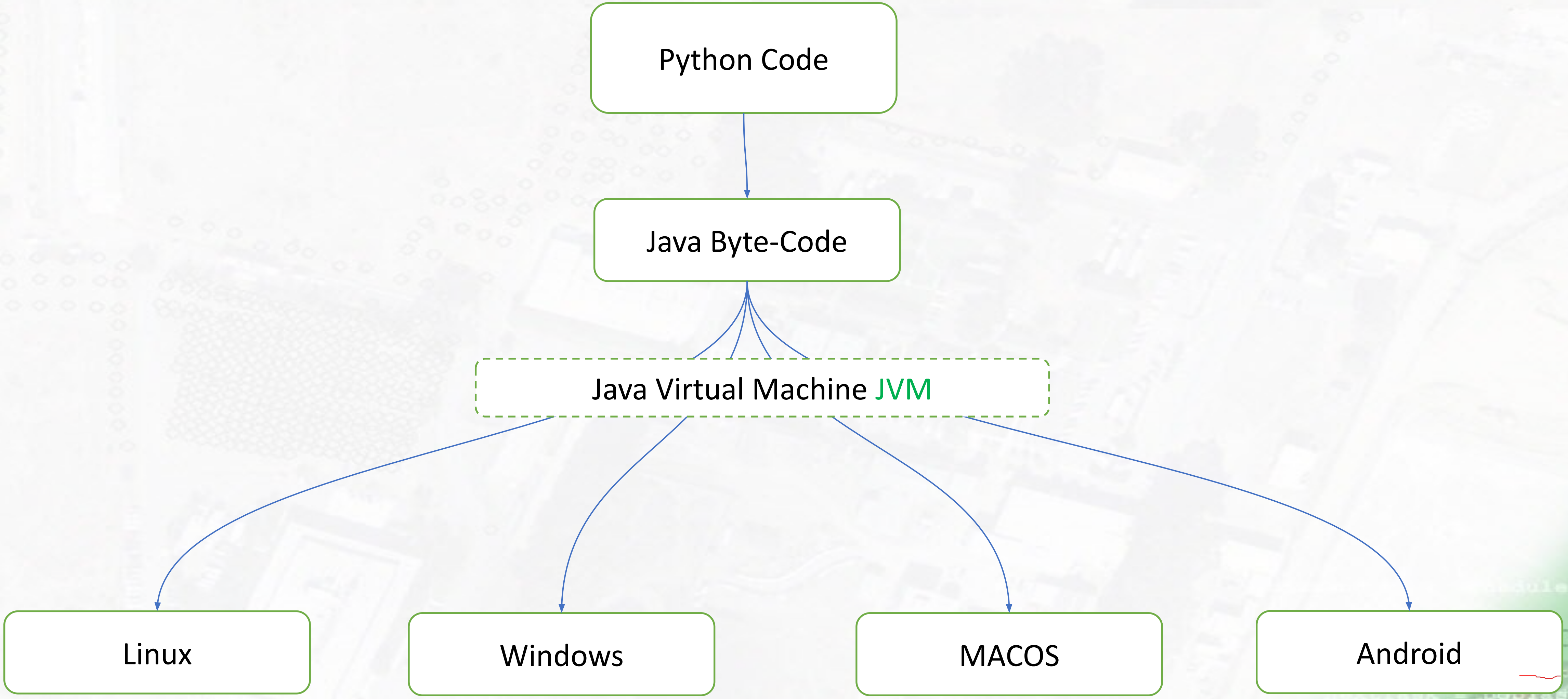

Не защищено
https://jenkins.diim-co.com/job/dim-war-deploy/
Jenkins
dim-war-deploy
Status
Changes
Собрать сейчас
Настройки
Удалить Pipeline
Full Stage View
Переименовать
Pipeline Syntax
Лог опроса Git
Builds
Фильтровать сборки...
Today
#611 18:15
#610 17:51
#609 16:33
#608 16:18
#607 16:05
#606 15:57
dim-war-deploy
изменить описание
Pipeline script from SCM
Stage View
Declarative: Checkout SCM
Build WAR
Deploy to WildFly
Verify Deployment
Average stage times:
(full run time: ~2min 47s)
1s
1min 11s
1s
1min 25s
#612
авг. 25
21:25
1 commit
1s
18s
#611
авг. 25
21:15
1 commit
995ms
50s
1s
1min 11s
#610
авг. 25
20:51
5 commits
1s
1min 4s
1s
1min 0s
#609
авг. 25
19:33
1 commit
1s
58s
1s
1min 21s
#608

Python Programming Language

Python Execution



Jython Execution



Compare

Basic Hello program in different languages

c- program

```c
#include <stdio.h>

int main(void) {
    printf("Hello World");
    return 0;
}
```

Python

```python
print "Hello world";
```

Java

```java
import java.io.*;
class GFG
{
    public static void main (String[] args)
    {
        System.out.println("Hello world");
    }
}
```

PHP

```php
<?php
    echo "Hello world";
?>
```

c++

```cpp
#include <iostream>

int main() {
    std::cout<<"Hello world";
    return 0;
}
```

فلسفة البايثون

```
import this
```

```
The Zen of Python, by Tim Peters

.Beautiful is better than ugly
.Explicit is better than implicit
.Simple is better than complex
.Complex is better than complicated
.Flat is better than nested
.Sparse is better than dense
.Readability counts
.Special cases aren't special enough to break the rules
.Although practicality beats purity
.Errors should never pass silently
.Unless explicitly silenced
.In the face of ambiguity, refuse the temptation to guess
.There should be one-- and preferably only one --obvious way to do it
.Although that way may not be obvious at first unless you're Dutch
.Now is better than never
.Although never is often better than *right* now
.If the implementation is hard to explain, it's a bad idea
.If the implementation is easy to explain, it may be a good idea
!Namespaces are one honking great idea -- let's do more of those
```

Why Learn Python

•Python continues to be one of the most widely used programming languages,

with particularly strong adoption in **AI, data science, and back-end development**.

•**Easy to learn → Powerful ecosystem → Useful across many IT careers**

•🌐 **Web Development :** Build websites, web services, APIs, and back-end systems.
•🤖 **AI & Machine Learning:** Develop intelligent systems, data models, computer vision, and automation.
•📊 **Data Science & Analytics:** Analyze data, create visualizations, and support data-driven decisions.
•🌐 **Networking & Automation:** Work with network protocols, sockets, system administration, and automated tasks.
•🔐 **Cybersecurity:** Create security tools, automate testing, analyze networks, and support penetration-testing workflows.
•🖥️ **Desktop & Scientific Applications:** Build desktop software, simulations, research tools, and engineering

Python ident

```
s='I Belongs to the first Level Blovk'
for i in s:
    i=i*2
    if i=='d':
        i=i*4
        print('I Belongs to the Third Level Blovk')
    print('I Belongs to the Second Level Blovk')
print('I Belongs to the first Level Blovk')
```

Level1

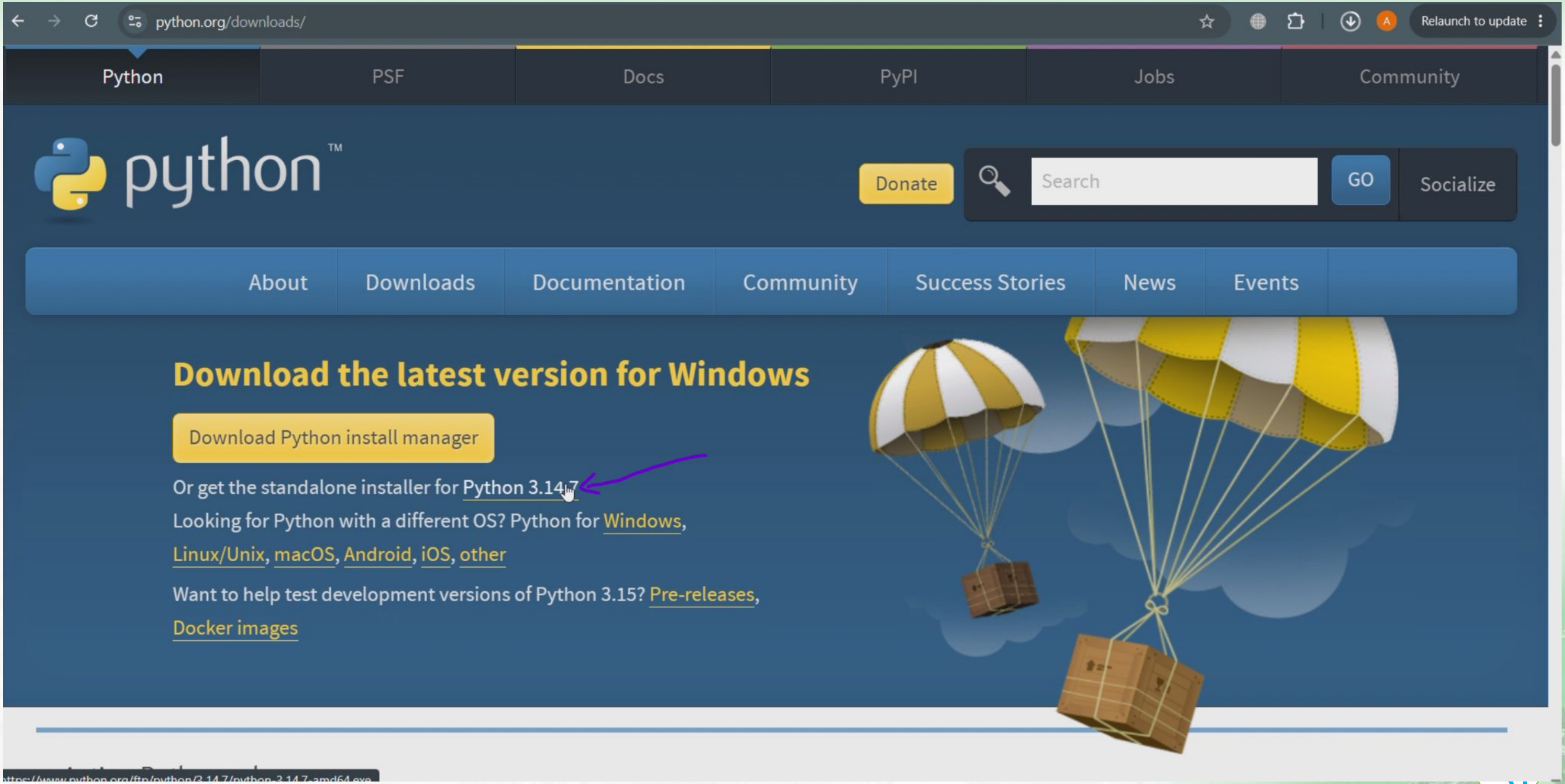
Level2

Level3

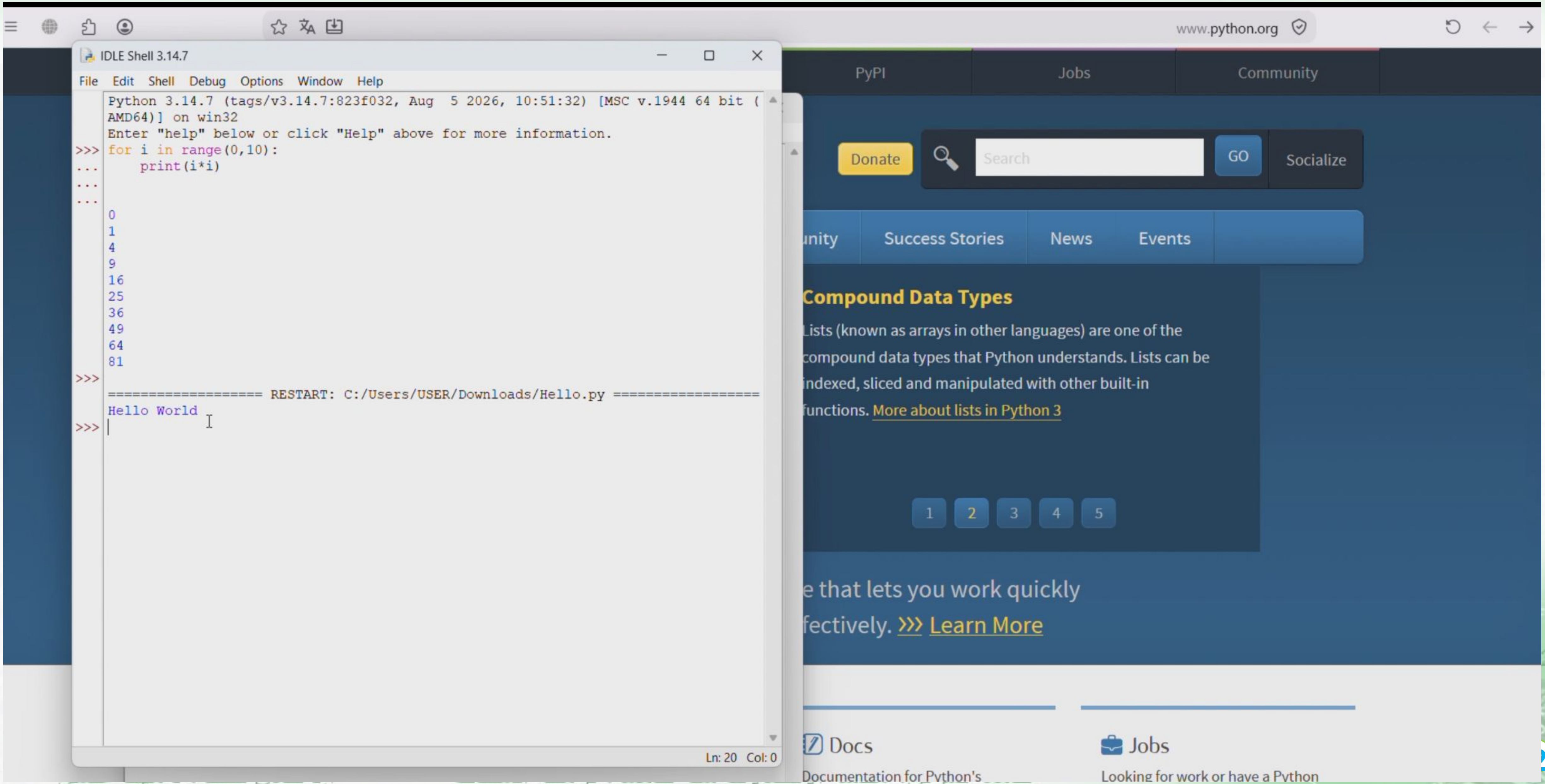
Level1

تنصيب البايثون

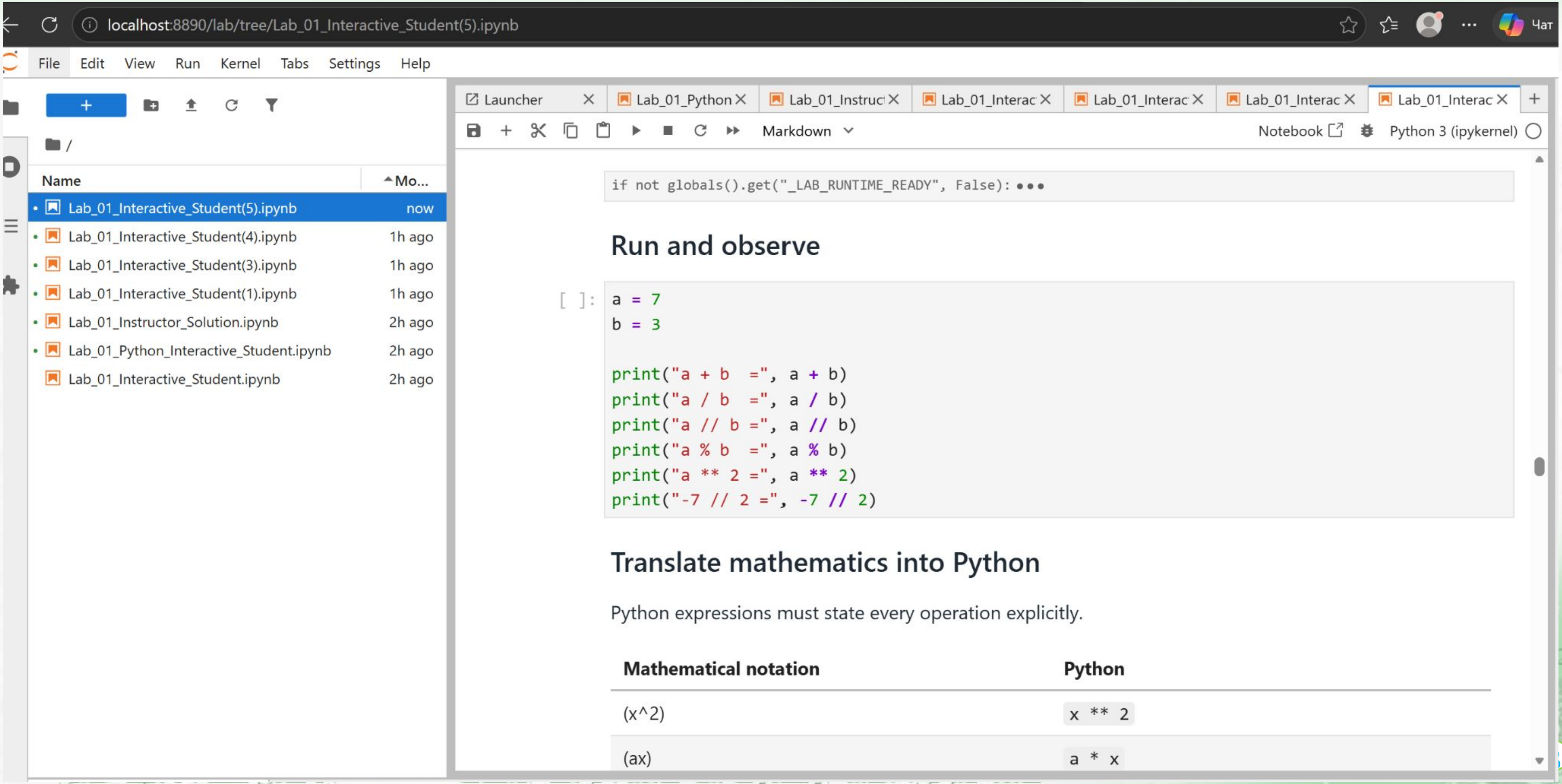
Install python



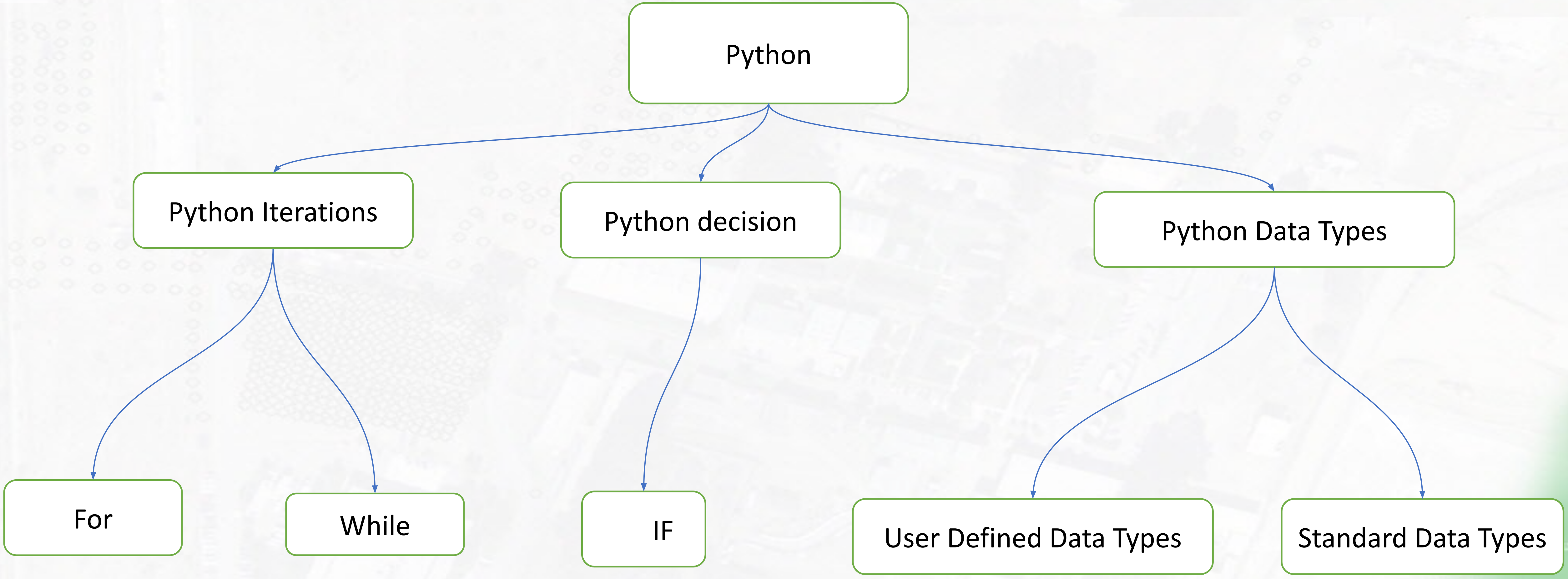
Default Python IDE “IDLE



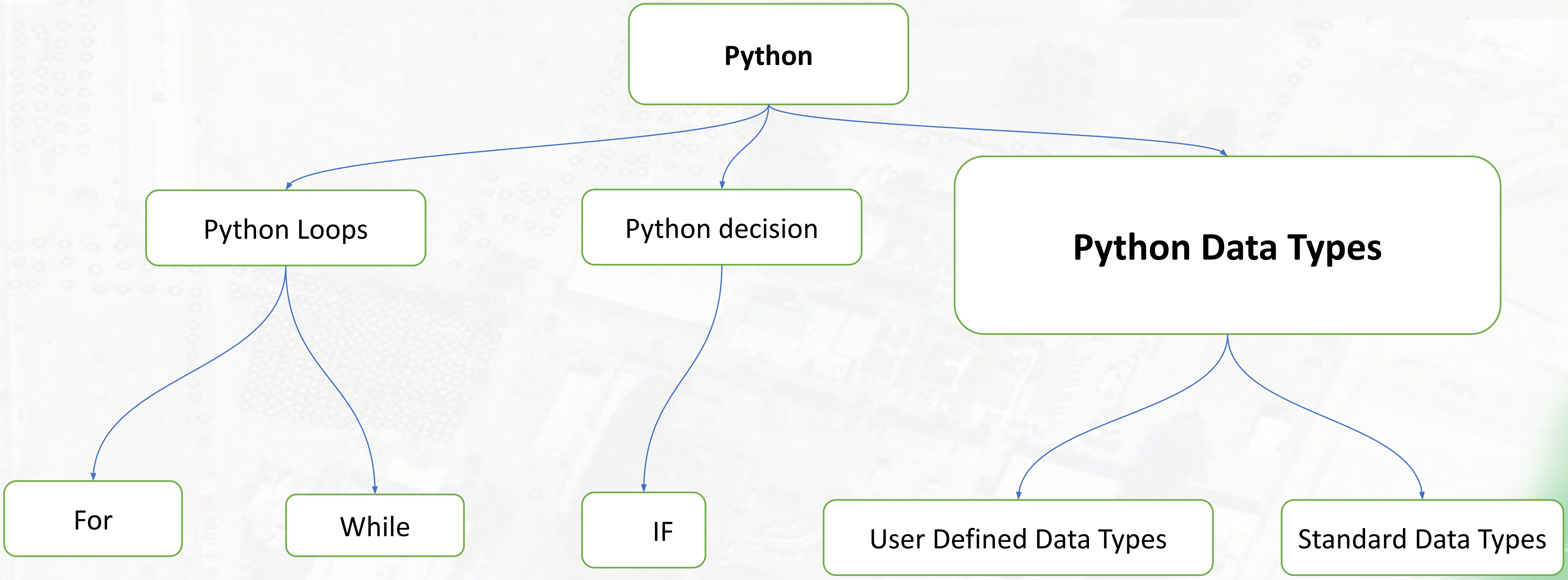
Install Jupyter LAB



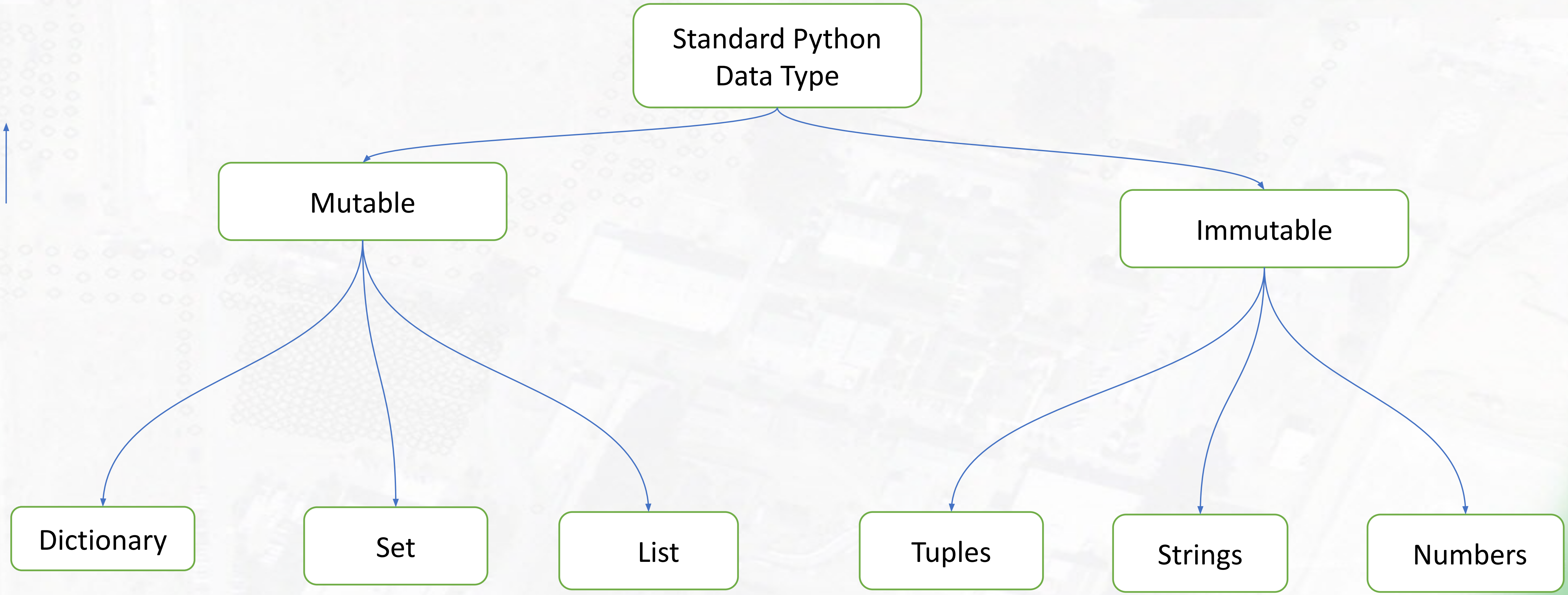
Python Data Structure



Python Data Type



Python Data Type



انماط البيانات الأساسية في بايثون

```
[92]: x=5.0

[93]: s='Network Programming'

[94]: l=['a','b',8,9]

[95]: t=('a','b',8,9)

[96]: s={'a','b',8,9}

[97]: dict={'a':25,'b':35}
```

Python Data Types is Dynamics

Built-in Function

```
x=5
```

```
type(x)
```

```
int
```

```
x='i was int, But now i become string'
```

```
type(x)
```

```
str
```

Built-in Function

```
[6]: dir(__builtins__)
```

```
[6]: ['ArithmeticError',
 'AssertionError',
 'AttributeError',
 'BaseException',
 'BlockingIOError',
 'BrokenPipeError',
 'BufferError',
 'BytesWarning',
 'ChildProcessError',
 'ConnectionAbortedError',
```

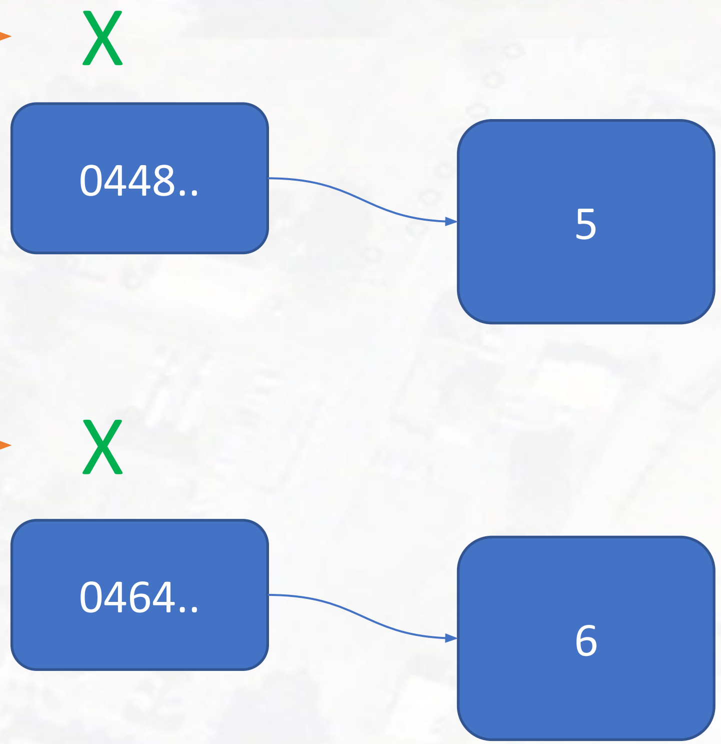
اظهار التوابع الخاصة بنمط بيانات محدد

```
[2]: s="hellow world"
[7]: dir(s)
  ...
```

IMMutable Data Types Integer

```
[8]:  x=5

[9]:  id(x)
[9]:  1477310448

[10]: x=5+1

[11]: id(x)
[11]: 1477310464

[12]: s='Good'

[14]: id(s)
[14]: 3948800

[15]: s=s+';'

[16]: id(s)
[16]: 3950496
```

ython 3... Saving com... Mode: Co...



Strings

```
#Strings
```

```
S='Network Programming'
```

```
S="Network Programming"
```

```
#Multi-line Strings
```

```
s='network programming languges\
     Python \
     java '
```

```
s="""
network programming languges
     Python
     java
  """
```

2020

strings

```
s='spam & eggs'
```

```
s[0]
```

```
's'
```

```
s[5]
```

```
'&'
```

```
s[-1]
```

```
's'
```

```
s[-4]
```

```
'e'
```

s	p	a	m		&		e	g	g	s
↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
0	1	2	3	4	5	6	7	8	9	10

s	p	a	m		&		e	g	g	s
↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
–11	–10	–9	–8	–7	–6	–5	–4	–3	–2	–1

Strings function & methods

```
[21]:  1  s='network programming'
```

```
[22]:  1  dir(s)
```

```
'lower',
'lstrip',
'maketrans',
'partition',
'replace',
'rfind',
'rindex',
'rjust',
'rpartition',
'rsplit',
```

List

```
l=[2,3]
id(l)
108304744
l.append(7)
id(l)
108304744
l
[2, 3, 7]
l.append([8,9])
l
[2, 3, 7, [8, 9]]
```

```
l
[2, 3, 7, [8, 9]]
id(l)
108304744
l.extend([12,15,20])
l
[2, 3, 7, [8, 9], 12, 15, 20]
id(l)
108304744
```

Split & join methods

```
108304744
```

```
s='a,b,c'
```

```
# convert string to list
```

```
l=s.split(',')
```

```
l
```

```
['a', 'b', 'c']
```

```
#**
```

```
s='a;b;c'
```

```
l=s.split(';')
```

```
l
```

```
['a', 'b', 'c']
```

```
l
```

```
['a', 'b', 'c']
```

```
#conver the list into string(join)
```

```
s=' '.join(l)
```

```
s
```

```
'a b c'
```

```
s='^'.join(l)
```

```
s
```

```
'a^b^c'
```

Copy list

```
# Copy List

l1=['a','b','c']

l2=l1

l2[0]='x'

l1

['x', 'b', 'c']

id(l2)

14928264

id(l1)

14928264
```

```
# Copy list

l1=['a','b','c']

l2=list(l1)

id(l1)

107872552

id(l2)

105337032

l1=['a','b','c']
l2=l1[:]

id(l1)

107905256

id(l2)

107873320
```

2020

Nested Lists

القوائم المتداخلة

```
# nested list
```

```
l=[3,[5,2,['first','second']],9,10]
```

```
l[1]
```

```
[5, 2, ['first', 'second']]
```

```
l[1][2]
```

```
['first', 'second']
```

```
l[1][2][0]
```

```
'first'
```

```
l[1][2][0][0]
```

```
'f'
```

List Comprehension

```
# List comperhension
```

```
# create even number list
```

```
l=[x for x in range (20) if x%2==0]
```

```
l
```

```
[0, 2, 4, 6, 8, 10, 12, 14, 16, 18]
```

```
# create odd number list
```

```
l=[x for x in range (20) if x%2!=0]
```

```
l
```

```
[1, 3, 5, 7, 9, 11, 13, 15, 17, 19]
```

Tuples

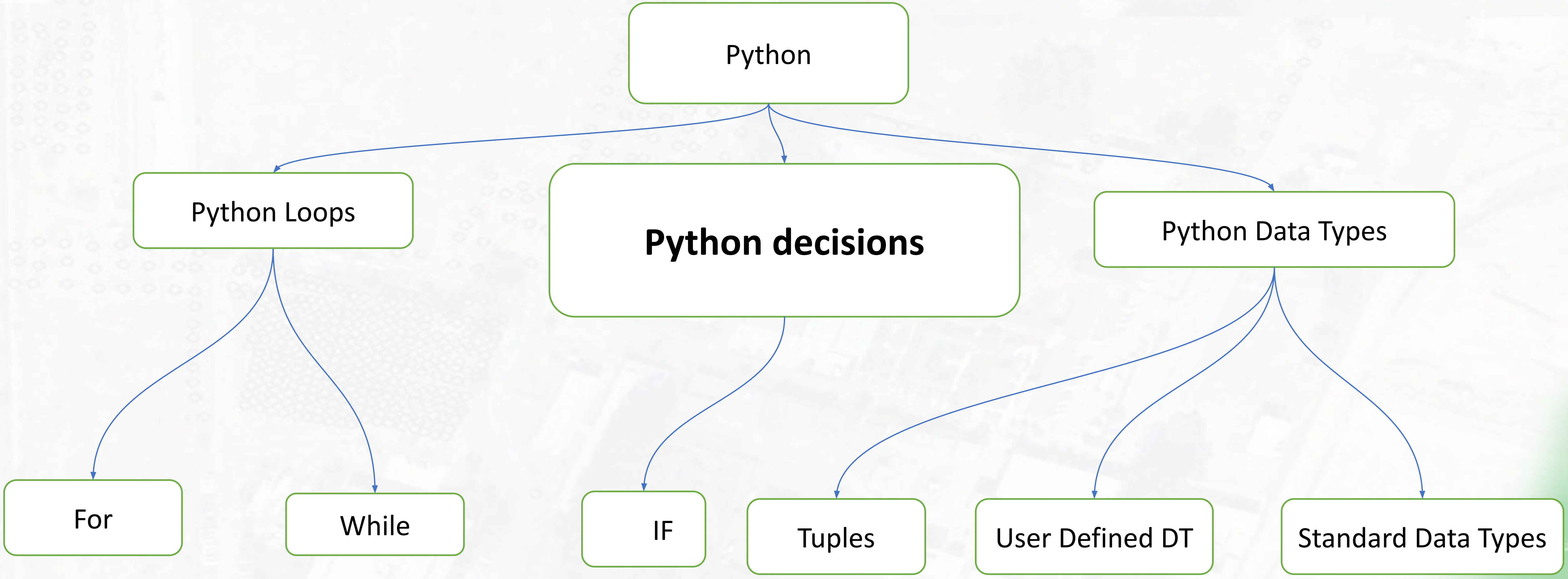
```
# Tuples
```

```
t=('a','b','c')
```

```
t[0]='k'
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
<ipython-input-41-f52af8b8506d> in <module>
----> 1 t[0]='k'

TypeError: 'tuple' object does not support item assignment
```

Python Data Type



```python
# two weay condition
```

```python
x = 50

if  x >= 50 :
    print('you passed the test')
else:
    print('Sorry, you Failed')
```

```
you passed the test
```

```python
# two weay condition
```

```python
x = 50

if   x > 50 :
     print('you passed the test')
elif x < 50 :
     print('Sorry, you Failed')
else:
     print('you Hardly passed')
```

```
you Hardly passed
```

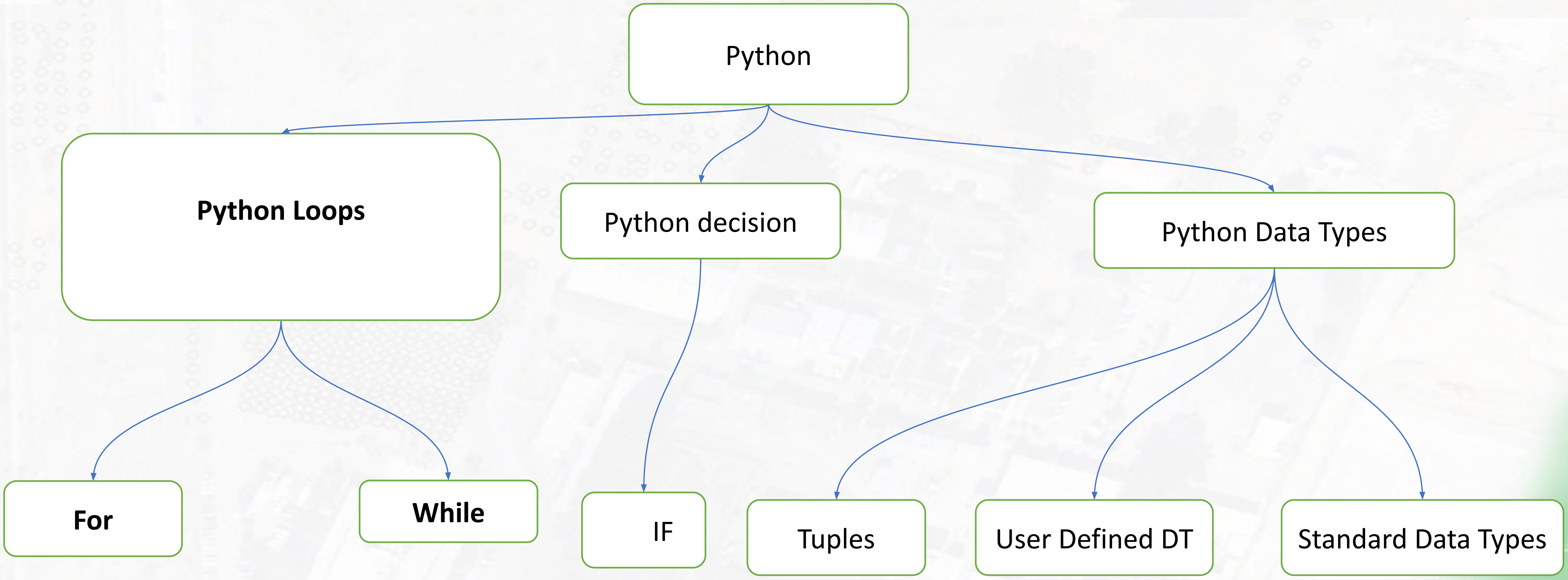
Nested Conditions

```
# nested  condition
```

```
x = 90

if isinstance(x, int):
     if x >= 50 :
          print('you passed the test')
     else :
          print('Sorry you failed')
else:
     print('please inter integer number
```

```
you passed the test
```

Python Data Type



Python Iteration

FOR LOOP

```
for i in range(1, 6):
    print(i)
```

WHILE LOOP

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

`for`

Best when iterating over an iterable such as a list, string, dictionary, or range.

`while`

Best when repetition should continue while a condition remains true.

PREDICT **Range check**

`range(1, 6)` produces 1, 2, 3, 4, 5 — the stop value 6 is excluded.

Do it In Python

```
********************
 ******************
  ****************
   **************
    ************
     **********
      ********
       ******
        ****
         **
```